HKUST Name: _____
MATH 2111 Matrix Algebra and Applications Student ID: _____
Sample C Fall 2026–2027 Midterm Test Lecture Section: _____

Total: 100 marks. *Closed book. No calculator or formula sheet. Answer all questions and show sufficient work for Questions 2–5.*

Qn. 1 (40 marks) Choose one correct option for each question. No justification is required. Each correct answer is worth 4 marks, with no deduction for a wrong answer.

(1) Which map $T : \mathbb{R}^2 \to \mathbb{R}^2$ is linear?

(A) $T(x_1, x_2) = (x_1 + 1, x_2)$
(B) $T(x_1, x_2) = (x_1 - x_2,\ 2x_1 + 3x_2)$
(C) $T(x_1, x_2) = (x_1 x_2, x_2)$
(D) $T(x_1, x_2) = (|x_1|, x_2)$

(2) Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ and $S : \mathbb{R}^3 \to \mathbb{R}^2$ have standard matrices A and B, respectively. The standard matrix of $S \circ T$ is

(A) AB
(B) $A^T B^T$
(C) BA
(D) $B^T A$

(3) Let A be a 4×3 matrix and B a 3×4 matrix such that $BA = I_3$. Which statement must be true?

(A) The columns of A are linearly independent.
(B) The columns of B are linearly independent.
(C) The map $x \mapsto Ax$ is onto $\mathbb{R}^4$.
(D) $AB = I_4$.

(4) Suppose $b \neq 0$, the system $Ax = b$ is consistent, x_p is one particular solution, and v_1, v_2 are linearly independent vectors satisfying

$$\mathrm{Nul}(A) = \mathrm{Span}\{v_1, v_2\}.$$

Which is the complete solution set of $Ax = b$?

(A) $x = x_p + s(v_1 + v_2)$
(B) $x = sv_1 + tv_2$
(C) $x = x_p + sv_1$
(D) $x = x_p + sv_1 + tv_2$, where $s, t \in \mathbb{R}$

(5) Let A be an invertible 3×3 matrix with $\det A = -4$. What is $\det(2A^{-1})$?

(A) -8
(B) $-\frac{1}{2}$
(C) -2
(D) 2

Qn. 1 (continued)

(6) Let $A = [\, a_1 \; a_2 \; a_3 \,]$ be an $m \times 3$ matrix, and let B be a $3 \times n$ matrix whose jth column is $[\, 2 \; -1 \; 3 \,]^T$. The jth column of AB is

(A) $2a_1 + a_2 - 3a_3$
(B) $2a_1 - a_2 - 3a_3$
(C) $a_1 - 2a_2 + 3a_3$
(D) $2a_1 - a_2 + 3a_3$

(7) Let $A = [\, a_1 \; a_2 \; a_3 \,]$. If $a_1 - 2a_2 + a_3 = 0$, which statement must be true?

(A) $[\, 1 \; 2 \; 1 \,]^T \in \mathrm{Nul}(A)$
(B) $[\, 1 \; -2 \; 1 \,]^T \in \mathrm{Nul}(A)$
(C) The rows of A are linearly independent.
(D) A has a pivot in every column.

(8) Evaluate

$$\det \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ 3 & 0 & 4 \end{bmatrix}.$$

(A) -5
(B) 5
(C) -11
(D) 11

(9) Let A be an $n \times n$ matrix satisfying $A^2 = 0$. Which statement must be true?

(A) $A = 0$.
(B) A is invertible.
(C) $\det A = 0$.
(D) $Ax = b$ is consistent for every $b \in \mathbb{R}^n$.

(10) Let A be a 4×4 matrix. If $Ax = b_0$ has exactly one solution for some $b_0 \in \mathbb{R}^4$, which statement must be true?

(A) $Ax = 0$ has a nonzero solution.
(B) The columns of A are linearly dependent.
(C) There exists $b \in \mathbb{R}^4$ for which $Ax = b$ is inconsistent.
(D) $Ax = 0$ has only the zero solution.

qn.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ans.										

Qn. 2 (14 marks) In $\mathbb{R}^4$, let

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

$$w_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -1 \end{bmatrix}.$$

Define $U = \text{Span}\{u_1, u_2, u_3\}$ and $W = \text{Span}\{w_1, w_2\}$.

(a) (6 marks) Find all vectors that belong to both U and W. Express your answer as the span of a linearly independent set.

(b) (4 marks) Show that $\{u_1, u_2, u_3, w_1\}$ is a basis of $\mathbb{R}^4$. Hence explain why every $z \in \mathbb{R}^4$ can be written as $z = u + w$ with $u \in U$ and $w \in W$.

(c) (4 marks) Suppose one such decomposition is $z = u_0 + w_0$. Describe *all* decompositions $z = u + w$ with $u \in U$ and $w \in W$. Is the decomposition unique?

Qn. 3 (16 marks) Let

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

A linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is required to satisfy

$$T(u_1) = e_1, \quad T(u_2) = e_2, \quad T(u_3) = \begin{bmatrix} 1 \\ p \\ q \end{bmatrix},$$

where e_1, e_2, e_3 are the standard basis vectors of $\mathbb{R}^3$.

(a) (4 marks) Determine all values of p and q for which such a linear transformation can exist. Justify your answer.

(b) (7 marks) For those values of p and q, find the standard matrix of *every* possible transformation T. You may set $T(e_3) = [\, a \; b \; c \,]^T$. Determine exactly which of these transformations are invertible.

(c) (5 marks) Among the transformations in part (b), determine all those satisfying $T(T(x)) = T(x)$ for every $x \in \mathbb{R}^3$. For each such transformation, find all fixed vectors satisfying $T(x) = x$.

Qn. 4 (15 marks) Define

$$R : \mathbb{R}^4 \to \mathbb{R}^2, \qquad R(x_1, x_2, x_3, x_4) = \begin{bmatrix} x_1 + x_3 \\ x_2 + x_3 + x_4 \end{bmatrix},$$

and

$$S : \mathbb{R}^2 \to \mathbb{R}^3, \qquad S(a, b) = \begin{bmatrix} a \\ b \\ a + b \end{bmatrix}.$$

Let $T = S \circ R$.

(a) (4 marks) Find the standard matrices $C = [R]$, $B = [S]$, and $A = [T]$. Verify that $A = BC$.

(b) (5 marks) Use the structure $T = S \circ R$ to find all $x \in \mathbb{R}^4$ such that $T(x) = 0$. Write the answer in parametric vector form.

(c) (6 marks) Describe all possible output vectors of T in two ways: as the span of a linearly independent set and as the solution set of one linear equation in y_1, y_2, y_3. Hence determine whether T is one-to-one and whether it is onto $\mathbb{R}^3$.

Qn. 5 (15 marks) For a real parameter α, define $T_\alpha : \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T_\alpha(x) = x + \alpha(x_1 + x_2 + x_3) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

(a) (5 marks) Find the standard matrix A_α of T_α. Using elementary row operations, compute $\det A_\alpha$.

(b) (6 marks) Determine all α for which T_α is invertible. For each such α, show that

$$T_\alpha \circ T_\beta = T_{\alpha+\beta+3\alpha\beta},$$

and hence find β such that $T_\beta = T_\alpha^{-1}$.

(c) (4 marks) Let $\alpha = -\frac{1}{3}$. Determine exactly which $b \in \mathbb{R}^3$ make the equation $T_{-1/3}(x) = b$ consistent, and give the complete solution set whenever it is consistent.

End of examination